Rejection and Accept-Reject Sampling: Recap

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Rejection sampling

The target is a candidate conditioned on an event: $Y = X \mid X \in A$.

- Draw X and **keep the draws that land in** A.
- The fraction kept is $P(X \in A)$. When A is rare, most draws are wasted.
- Examples: $X \mid X$ odd, a truncated normal, points in the unit disk.

Truncated normal, $Y = Z \mid a \le Z \le b$:

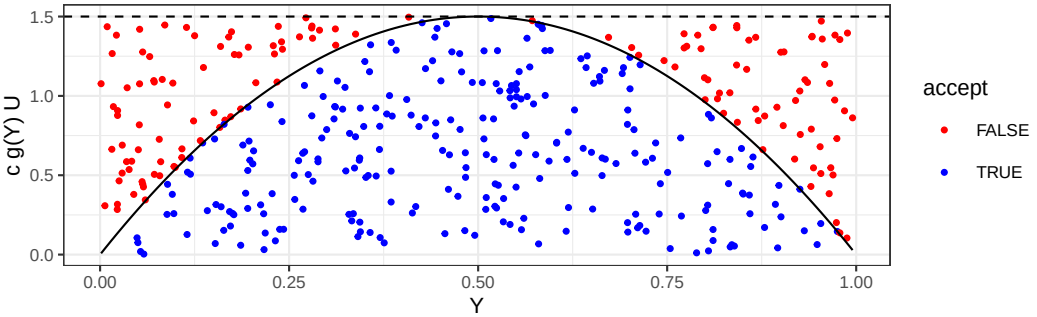
$$F_Y(y) = \frac{\Phi(y) - \Phi(a)}{\Phi(b) - \Phi(a)}, \qquad Q_Y(p) = \Phi^{-1}(\Phi(a) + p\,[\Phi(b) - \Phi(a)]).$$

Check the kept draws with a QQ plot against Q_Y.

Accept-reject

Target density f, candidate density g, and a constant c with $c\,g(x) \geq f(x)$ wherever $f(x) > 0$.

1. Draw $Y \sim g$ and $U \sim U(0,1)$.
2. **Accept** $X = Y$ if $U \leq f(Y)/(c\,g(Y))$; otherwise reject.



Beta(2, 2) under a uniform envelope with $c = 1.5$: keep the points below f.

Why it works

The output is a conditional distribution: $V = Y \mid U \le f(Y)/(c\,g(Y))$.

$$P(V \le x) = \frac{P(Y \le x,\ U \le f(Y)/(c\,g(Y)))}{P(U \le f(Y)/(c\,g(Y)))} = \frac{\frac{1}{c}F(x)}{\frac{1}{c}} = F(x)$$

- The numerator step $\int_0^{f(y)/(cg(y))} du = f(y)/(c\,g(y))$ needs the ratio to be **at most 1**. That is why $c\,g \ge f$.
- The denominator is the **acceptance rate**, $1/c$.

Choosing g and c

- The best c is $\max_x f(x)/g(x)$. Find it by calculus, or evaluate the ratio on a grid.
- A c that is too large is still correct but wastes candidates.
- From m candidates about m/c are kept, so $\mathrm{Var}\left(\bar{X}\right) \approx c\,\mathrm{Var}\left(X\right)/m$.
- A g closer in shape to f gives a smaller c. For the truncated normal on $[0,1]$:

envelope	c	rejected
$U(0,1)$	1.17	14%
$\frac{2}{3}(2-y)$	1.06	6%

The uniform envelope's estimate has about 10% more variance.

When accept-reject fails

Tails too light. For $f(x) = 0.25\,\phi(x+2) + 0.75\,\phi(x-1)$ and $g = \phi$, $f(x)/\phi(x) \to \infty$, so no c exists.

- g's tails must be at least as heavy as f's.
- The Cauchy works: its tails fall like $1/x^2$.
- A c found by evaluating on a grid only covers the grid. Check that the ratio is bounded outside it.